RADICAL RULES

Which rule holds for which index

Odd roots take any radicand; even roots ask for signs first.

✓ ALWAYS · ◐ SIGN CONDITION · ✕ NO REAL VALUE

RULE	Even index	Odd index
Product rule $\sqrt[n]{a}\,\sqrt[n]{b} = \sqrt[n]{ab}$	◐ SIGN CONDITION both factors ≥ 0	✓ ALWAYS any radicands
Quotient rule $\sqrt[n]{a}/\sqrt[n]{b} = \sqrt[n]{a/b}$	◐ SIGN CONDITION numerator ≥ 0, denominator > 0	✓ ALWAYS any radicands
Power rule $(\sqrt[n]{a})^m = \sqrt[n]{a^m}$	◐ SIGN CONDITION base ≥ 0	✓ ALWAYS any base
Negative radicand $\sqrt[n]{-a}$	✕ NO REAL VALUE $\sqrt{-4}$ needs complex numbers	✓ ALWAYS $-\sqrt[n]{a}$, e.g. $\sqrt[3]{-27} = -3$

All three rules also need the same index on every radical: $\sqrt{a} \cdot \sqrt[3]{b}$ has no single-radical form until both are written as rational exponents.